

Vue.js Props Practice Lab

1. Executive Project Overview

The **Vue.js Props Practice Lab** is a high-fidelity educational architecture designed specifically for beginner to intermediate frontend students. The core pedagogical purpose of this exercise is to guide students through the strategic deconstruction of a monolithic interface into clean, reusable child components while establishing scalable parent-to-child data flows via `props`.

Target Student Knowledge Prerequisites:

- Vue 3 Composition API structure (specifically utilizing `ref()` primitives)
- Basic Single-File Component (SFC) registration and templates
- Dynamic attribute data-binding (`v-bind` or `:prop` shorthand)
- Standard declarative event handling (`v-on` or `@name`)

2. Component Hierarchy & System Architecture

Students are tasked with taking an un-factored page layout and slicing it cleanly into the hierarchical framework outlined below. In this laboratory design, `App.vue` serves as the primary data orchestrator (holding global configuration state).

App.vue (Root Component / Orchestrator)

- `Header.vue`
- `CourseInfo.vue`
- `StatsSection.vue` → Loop of `StatCard.vue`
- Main Work Area (Grid Split):
 - `StudentProfile.vue`
 - `AssignmentList.vue` → Loop of `AssignmentCard.vue`
 - `ActivityTimeline.vue` → Loop of `ActivityItem.vue`
- `Footer.vue`

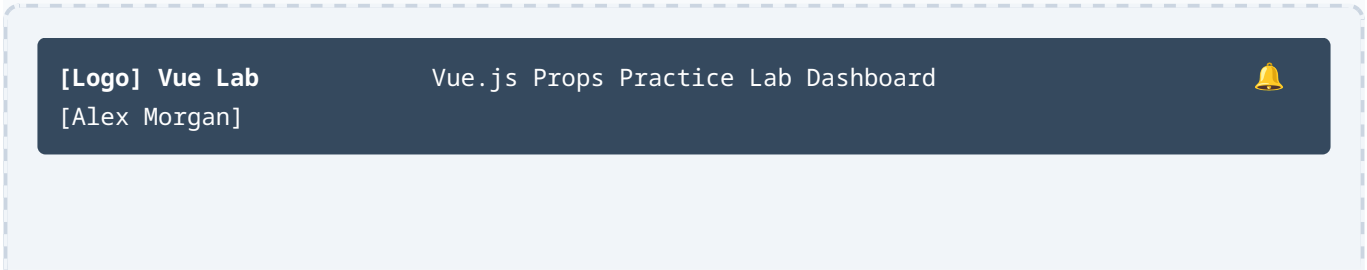
3. Props Mapping & Data Typings Specification

To successfully compile the application, students must implement strict type-checking validation inside their `defineProps()` configurations. Below is the mandatory blueprint specification table:

Target Component	Prop Property Name	Data Type	Realistic Mock Value Snippet
StatCard.vue	title	String	"Lessons Completed"
	value	String Number	12
	icon	String	"book-open"
AssignmentCard.vue	title	String	"Composition API Quiz"
	dueDate	String	"June 15"
	status	String	"pending" "completed"
ActivityItem.vue	activityName	String	"Submitted Exercise"
	date	String	"2 mins ago"
	type	String	"submit" "quiz" "unlock"
StudentProfile.vue	avatar	String	"https://placeholder.co/150"
	fullName	String	"Alex Morgan"
	email	String	"alex.m@edu.com"
	batch	String	"2026-A"
	track	String	"Frontend Development"
CourseInfo.vue	courseName	String	"Frontend Frameworks (Vue 3)"
	instructor	String	"Prof. Evans"
	progress	Number	60

4. Blueprint UI Mockup Blueprint (1440px Desktop Perspective)

The blueprint diagram below represents the unified visual interface configuration that students will aim to reproduce programmatically using their decoupled design pattern.



Course: Frontend Frameworks (Vue 3) | Instructor: Prof. Evans | Progress: [██████░░░░] 60%

Lessons Completed
12 / 20

Exercises Done
8 / 10

Current Score
92%

Attendance Rate
95%

[Avatar Image]

Alex Morgan
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Batch: 2026-A
Track: Frontend

Upcoming Assignments

✓ Composition API Quiz
[COMPLETED]

⌚ Props Lab Exercise
[PENDING - JUNE 15]

Recent Activities

● Submitted Exercise
2 mins ago

● Quiz Completed
1 hour ago

5. Multi-Device Responsive Execution Rules

Tablet Topology (768px - 1024px)

- The main 3-column layout collapses to an optimized multi-row pattern.
- The 4 stats tracking indicators rearrange dynamically into a balanced 2x2 grid format.

Mobile Topology (375px - 425px)

- The interface strictly maps directly into a linear, unified vertical layout column.
- Header element texts drop down contextually to hide long text strings, prioritising the profile badge and essential navigation markers.
- Statistical metrics cards swap cleanly into a horizontal scroll strip (`overflow-x: auto`) to preserve maximum interactive workspace layout.

6. Classroom Implementation Blueprint Guide

Teacher Tip: Provide students with a repository containing a complete monolithic `App.vue` page including styles and reactive definitions inside an array collection block. Instruct students to carefully harvest specific HTML fragments, extract them out cleanly into standard single components files, configure the property declaration rules using `defineProps()`, and pass the active mock configuration data states downwards cleanly.